

Abstract

Human-in-the-Loop Communication: Repositioning Communication Pedagogy in AI-Driven Education

The growing use of artificial intelligence (AI) tools in higher education is changing how students produce written work and how learning is assessed. As generative AI systems are now able to produce coherent academic texts, writing on its own is no longer a dependable indicator of student understanding. This development calls for a reconsideration of how learning is demonstrated and evaluated.

This paper introduces the concept of *human-in-the-loop communication* as a pedagogical framework that places communication, not writing alone, at the centre of learning validation. The framework presents learning as a cyclical process involving three interconnected elements: human cognitive foundations (clarity, structure, audience awareness), AI-supported practice (drafting, structuring, feedback, idea development), and human validation (speaking, discussion, and reflection). Within this model, AI functions as a support tool in the learning process, while human communication remains the primary means through which understanding is made visible and assessed.

Drawing on classroom practices in applied sciences and international business education, the paper demonstrates how communication-based assessment formats, such as presentations, peer discussions, and opponent-based feedback, can better capture students' reasoning, adaptability, and depth of understanding. These approaches reflect the realities of AI-supported learning environments while maintaining academic integrity and meaningful assessment.

The paper contributes to current discussions on AI in education by offering a practical and transferable framework for redesigning teaching and assessment. It argues that the future of education lies not in resisting AI, but in designing learning environments that distinctly emphasise human communicative competence.

Keywords: Human-in-the-loop communication; artificial intelligence in education; communication pedagogy; assessment in higher education; generative AI; student learning; oral communication; applied sciences education